Report for

E-Commerce Sales Analysis Full Project

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Executive Summary

This report analyzes the sales and profit performance of an e-commerce store using the Sample Superstore dataset. The analysis identifies trends across months, categories, sub-categories, and customer segments.

Objective

The objective of this analysis is to study sales and profit patterns, identify top and low-performing areas, and provide insights to improve business performance.

Key Findings

- Highest Sales Month: **November**
- Lowest Sales Month: **February**
- Highest Profit Month: **December**
- Best Performing Category: **Technology**
- Worst Performing Category: **Furniture**

Dataset Description

- Dataset Name: Sample Superstore
- Number of Records: **9994**

Features Included: Order Date, Sales, Profit, Category, Sub-Category, Segment, Region, etc.

Monthly Sales Analysis

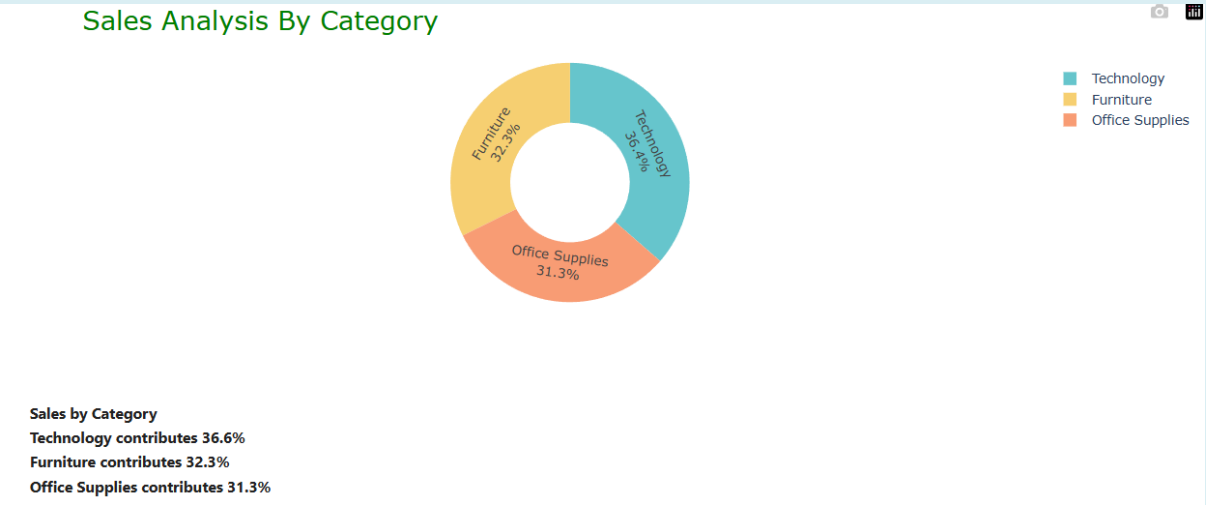
Highest Sales

Lowest Sales

- Month with Highest Sales: **November (352,461.07)**
- Month with Lowest Sales: **February (59,751.25)**

Observations: Sales are highest during the end of the year due to festive demand and lowest at the beginning of the year.

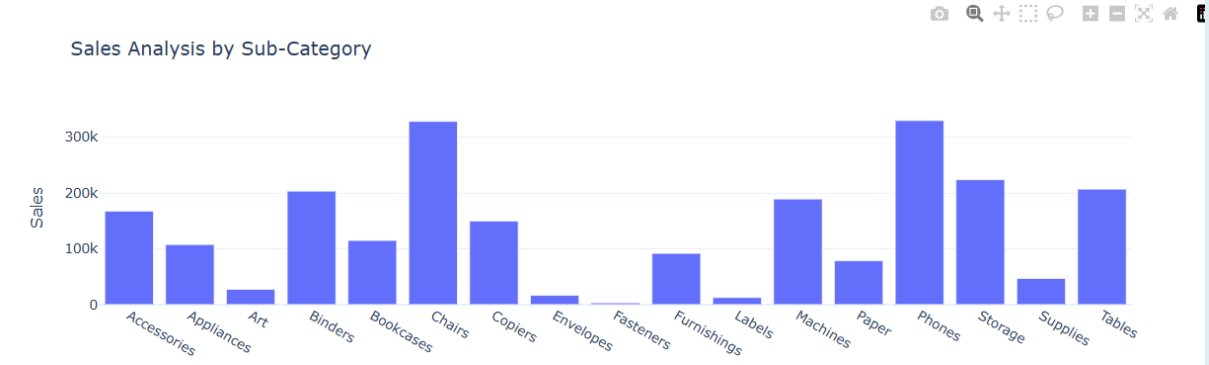
Sales by Category



- Highest Sales Category: **Technology (836,154.03)**
- Lowest Sales Category: **Office Supplies (719,047.03)**

Observations: Technology products generate maximum revenue due to higher prices.

Sales by Sub-Category



- Highest Sales Sub-Category: **Phones (330,007.05)**
- Lowest Sales Sub-Category: **Fasteners (3,024.28)**

Observations: Phones dominate sales, while small utility items contribute minimally.

Monthly Profit Analysis

Highest Profit

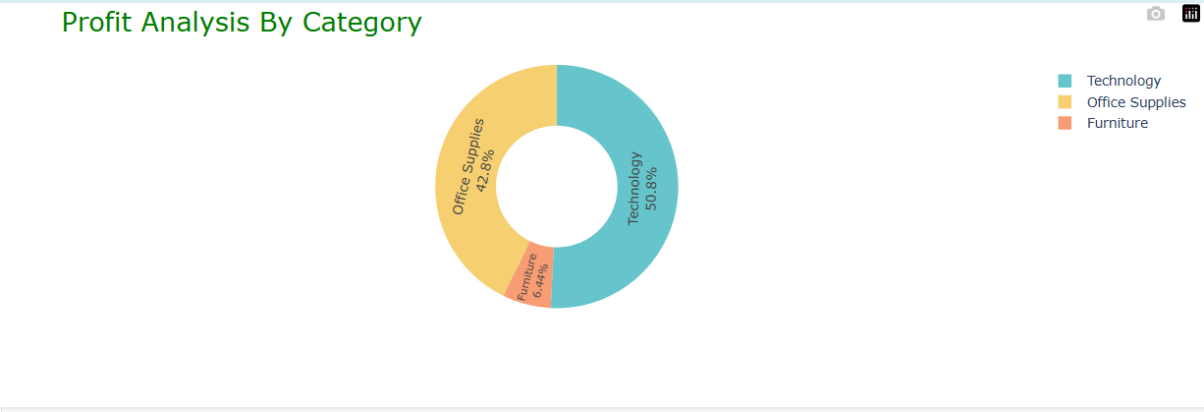
Lowest Profit

- Month with Highest Profit: **December (43,369.19)**
- Month with lowest profit : January (9134.446)

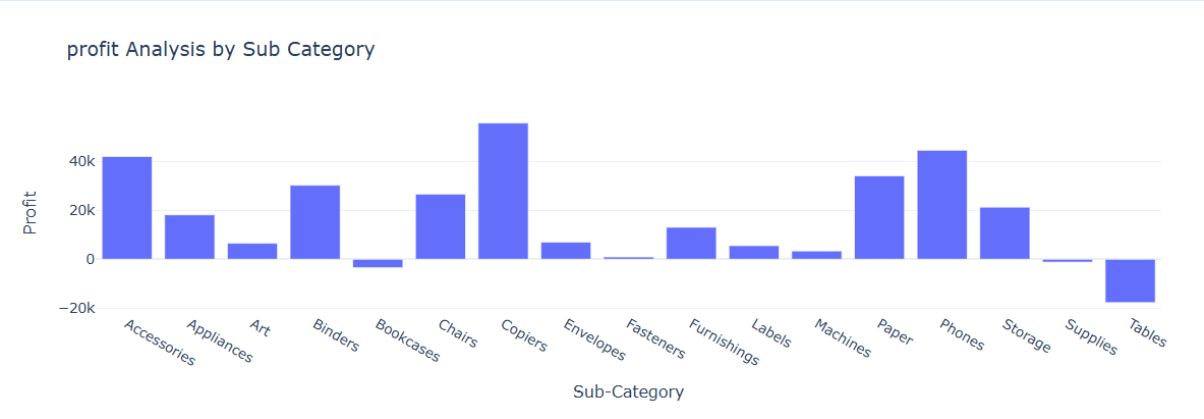
Observations: Profit peaks in December due to increased sales volume.

Profit by Category and Sub-Category

Profit Analysis By Category

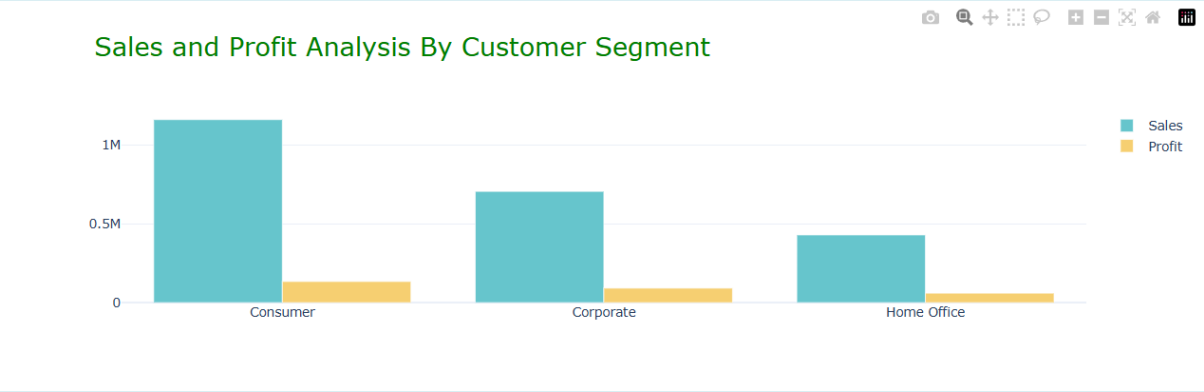


Profit Analysis By Sub-Categories



- Most Profitable Category: **Technology (145,454.95)**
- Least Profitable Category: **Furniture (18,451.27)**
- Most Profitable Sub-Category: **Copiers (55,617.82)**
- Least Profitable Sub-Category: **Tables (-17,725.48)**
- Observations: Furniture category shows weak profitability due to heavy discounts and costs.

Sales and Profit by Customer Segment



- Highest Sales Segment: **Consumer (1,161,401)**
- Highest Profit Segment: **Consumer (134,119.21)**
- Observations: Consumer segment contributes the most to both sales and profit.

Sales to Profit Ratio Analysis

Sales to Profit Ratio = Sales / Profit

- Consumer: **8.66**
- Corporate: **7.68**
- Home Office: **7.13**
- Best Performing Segment: **Home Office**

Observations: Lower ratio indicates higher efficiency; Home Office performs best.

Key Insights

- Sales and profit peak in November and December
- Technology is the strongest category in both sales and profit
- Furniture, especially Tables, results in losses

Recommendations

- Focus marketing efforts on Technology products
- Reduce discounts on Furniture items
- Promote high-profit products like Copiers and Phones

Conclusion

The analysis highlights important sales and profit trends, showing that Technology drives revenue while Furniture needs improvement. Seasonal trends also play a key role in performance.

Tools Used

- Python (Pandas, Matplotlib, plotly)
- SQL
- Jupyter Notebook